# Factorising and Expanding Brackets

## Fluency

### Expand

**Worked example:** Expand $2(x + 5)$. Multiply everything inside the brackets by 2: $2(x + 5) = 2 \times x + 2 \times 5 = 2x + 10$.

(a) $3(x + 4)$

(b) $5(2x - 3)$

(c) $-2(x + 6)$

(d) $4(3x - 5)$

(e) $-(x - 7)$

(f) $x(x + 3)$

(g) $2x(x - 5)$

(h) $-3x(2x + 1)$

### Factorise

**Worked example:** Factorise $8x + 20$. The HCF of $8x$ and 20 is 4, so $8x + 20 = 4(2x + 5)$.

(a) $4x + 12$

(b) $6x - 15$

(c) $x^2 + 5x$

(d) $3x^2 - 9x$

(e) $10x + 25$

(f) $2x^2 + 8x$

(g) $9x - 6$

(h) $9x^2 - 6x$

## Problem Solving

### Expand and simplify

**Worked example:** Expand and simplify $2(x + 4) + 3(x - 1)$:

$$2(x+4)+3(x-1) = 2x+8+3x-3 = 5x+5.$$

**Worked example:** Expand $(x+1)(x+6)$ by multiplying each term in the first bracket by each term in the second, then collecting like terms:

$$(x+1)(x+6) = x^2+6x+x+6 = x^2+7x+6.$$

(a) $3(x+2)+2(x-5)$

(b) $4(2x-1)-3(x+2)$

(c) $(x+3)(x+4)$

(d) $(2x-1)(x+5)$

(e) $(x-6)(x-2)$

### A step further

(a) A rectangle has length $(x + 5)$ and width $(x - 2)$. Write an expression for its area, expanded and simplified.

(b) The length of a rectangle is $2x-3$ and its width is $x$. The area is $2x^2 - 3x$. Factorise this area expression to check it matches the length and width given.

### Factorise fully

*Find the HCF of every term — it can include more than one letter, and a power of a letter.*

**Worked example:** Factorise $12x^4y+18x^2y^3$. The HCF of 12 and 18 is 6; the HCF of $x^4$ and $x^2$ is $x^2$; and $y$ is common to both terms, giving HCF $6x^2y$:

$$12x^4y+18x^2y^3 = 6x^2y(2x^2+3y^2).$$

(a) $18x^5y^2+12x^7$

(b) $8a^3b^2-20a^2b^4$

(c) $15x^3y-9x^2y^2+6xy^3$

(d) $14p^4q^3+21p^2q^5$

(e) $24m^6n^2-16m^4n^5+8m^3n^3$

### Solve

*Expand the brackets first, then solve for $x$.*

**Worked example:** Solve $4(x - 1) = 16$. Expand the bracket: $4x - 4 = 16$. Add 4 to both sides: $4x = 20$. Divide both sides by 4: $x = 5$.

(a) $3(x+5) = 27$

(b) $2(3x-4) = 5x+1$

(c) $5(x-2)-3(x+1) = 7$

## Reasoning

(a) Write 98 as $(100 - 2)$. Use expanding to calculate $7 \times 98$.

(b) Use factorising to quickly calculate $37 \times 8 + 37 \times 2$.

(c) A fruit stall sells 24 apples at \$2 each and 24 oranges at \$3 each. Use factorising to calculate $24 \times 2 + 24 \times 3$ in one step.

(d) Write 203 as $(200 + 3)$. Use expanding to calculate $6 \times 203$.

(e) Write 104 as $(100 + 4)$. Use expanding to calculate $9 \times 104$.

(f) Use factorising to calculate $85 \times 12 - 85 \times 2$.

## Enrichment

**Factorising Quadratics**

*A quadratic $x^2 + bx + c$ factorises as $(x + p)(x + q)$, where $p$ and $q$ are two numbers that* multiply *to give $c$ and* add *to give $b$. A difference of two squares, $a^2 - b^2$, always factorises as $(a - b)(a + b)$.*

**Worked example (sum and product):** Factorise $x^2 + 5x + 6$. Two numbers that multiply to 6 and add to 5 are 2 and 3, so $x^2 + 5x + 6 = (x + 2)(x + 3)$.

**Worked example (difference of two squares):** Factorise $x^2 - 25$. Since $x^2 - 25 = x^2 - 5^2$, this factorises as $x^2 - 25 = (x - 5)(x + 5)$.

(a) $x^2 + 7x + 10$

(b) $x^2 - 3x - 18$

(c) $x^2 - 16$

(d) $x^2 + 2x - 15$

(e) $x^2 - 49$

**Inequalities**

*Inequalities behave like equations, except multiplying or dividing by a negative number reverses the inequality sign.*

**Worked example:** Solve $2(x-3) < 10$.

$$2(x-3) < 10 \;\Rightarrow\; 2x - 6 < 10 \;\Rightarrow\; 2x < 16 \;\Rightarrow\; x < 8.$$

(Expand the bracket, add 6 to both sides, then divide both sides by 2.)

**Worked example (dividing by a negative):** Solve $-2x + 5 > 1$.

$$-2x + 5 > 1 \;\Rightarrow\; -2x > -4 \;\Rightarrow\; x < 2.$$

(Subtract 5 from both sides, then divide both sides by $-2$ — since we divided by a negative number, the $>$ sign flips to $<$.)

(a) Solve: $3(x+2) > 15$

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(b) Solve: $2(x-5) \leq 4(x+1)$

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(c) Solve: $-3(x+1) < 2(x-4)$

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(d) Solve the double inequality: $2 < 3(x-1) \leq 11$

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(e) A taxi charges a \$4 call-out fee plus \$3 per kilometre travelled. Write an inequality for the total cost of a trip of $k$ kilometres, then solve it to find how far you could travel and keep the cost under \$40.

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